(12) INTERNATIONAL APPLICATION PUBLISHED UNDER THE PATENT COOPERATION TREATY (PCT)

(19) World Intellectual Property Organization
International Bureau

(43) International Publication Date
15 March 2018 (15.03.2018)

WIPO | PCT

(10) International Publication Number
WO 2018/046399 A1

(51) International Patent Classification:
*G06Q 50/04* (2012.01)

(21) International Application Number:
PCT/EP2017/071961

(22) International Filing Date:
01 September 2017 (01.09.2017)

(25) Filing Language: English

(26) Publication Language: English

(30) Priority Data:
16187482.1 06 September 2016 (06.09.2016) EP

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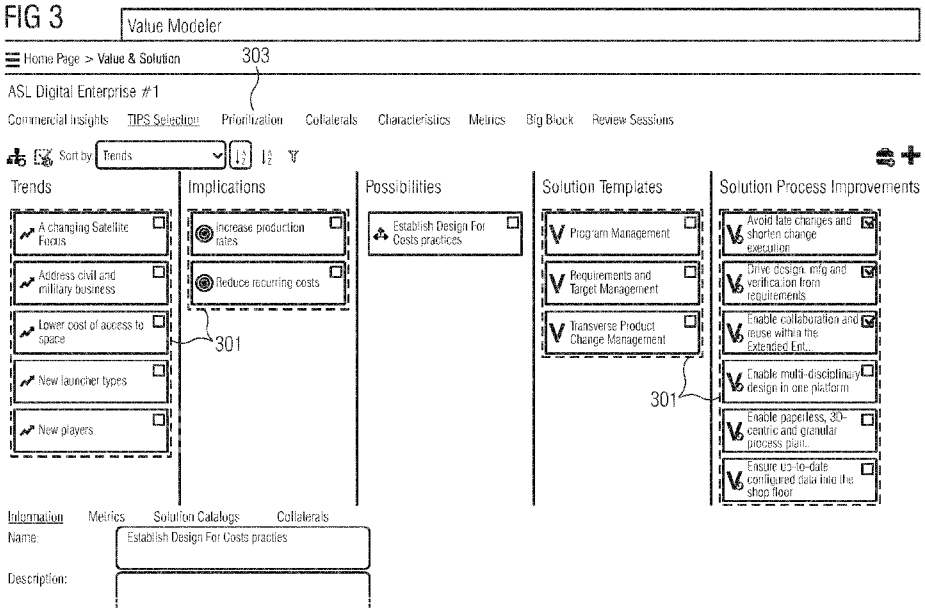
(81) Designated States *(unless otherwise indicated, for every kind of national protection available)*: AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA,

(54) Title: GENERATING A DIAGRAM OF INDUSTRY SOLUTIONS




(57) Abstract: A process and a system for generating a specific diagram of industry solutions, comprising a repository (401); the repository comprising a set of basic entities (201) representing a set of industry specific indicators; providing, within the repository (401), a set of industry solution entities (202) representing enablers of process improvements; linking at least two basic entities (201) to generate at least one relationship network (301); linking the at least one relationship network of basic entities to at least one industry solution entity (202); selecting a specific subset of basic entities (201) relevant for a specific customer;assigning to at least one basic entity (201) of the specific basic entity subset a basic value representing the relevance of the basic entity for the specific customer;generating a specific subset of industry solution entities (202) as a result of the selected basic entities in their relevant relationship network; selecting relevant industry solution entities from the specific solution entity subset;generating a specific diagram of industry solutions based on

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[Continued on next page]

SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.

**(84) Designated States** *(unless otherwise indicated, for every kind of regional protection available)*: ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

**Published:**

— *with international search report (Art. 21(3))*

the selected relevant industry solution entities of the previous step.

# Generating a diagram of industry solutions

The present invention relates to a method and to a system for generating a specific diagram of industry solutions according to the preamble of claims 1 and 5 respectively.

In order to improve their manufacturing processes, industrial manufacturers make us of industrial solutions for meeting their business needs and challenges.

Nowadays, for industrial manufacturers, the landscape of industry solutions provided by industrial suppliers is becoming more and more complex.

Thus, industrial manufacturers require guidance in order to understand where and how to start to improve manufacturing processes by their industry solution suppliers.

Software and hardware solution suppliers are faced with the challenge to provide solutions, hardware and/or software, able to improve the production and process of industrial manufacturers.

Such solutions need to be brought efficiently and timely into market to fit the customers' need.

Such solutions need to address specific customer requirements, which are industry specific but are also required to be tailored to the specific needs and challenges of a given customer and industrial facility.

Nowadays, unfortunately, industry solution suppliers often collect manually industry specific information relevant for a customer pursuit, with the support of traditional office documents e.g. slide decks, spreadsheets, etc.

For example, Siemens Industry Software makes use of a documentation based on "TIPS" framework and on pursuit specific documents e.g. big block diagrams, reports and dashboards.

The chosen descriptive term "TIPS" is an acronym for Trends, Implication, Possibilities and Solutions. The Trends are impacting the manufacturer business. The Implications are provided to the manufacturer by such trends. The Possibilities refer to how the manufacturer capitalizes on these implications and where the greatest risk is. The Solutions are the value contributions to the manufacturer's business.

Improved techniques for timely providing industry solutions to fulfil the customer requirements are desirable.

It is aim of the present invention to have a technique for automatic generating of a customer-specific diagram of industry solution according to the customer requirements and that proves consistent with the generic industry specific solution product portfolio of an industry supplier.

The aforementioned aim is achieved by a process and a system for generating a specific diagram of industry solutions, comprising:

a) providing a repository;

b) providing, within the repository, a set of basic entities representing a set of industry specific indicators;
c) providing, within the repository, a set of industry solution entities representing enablers of process improvements;
d) linking at least two basic entities to generate at least one relationship network;
e) linking the at least one relationship network of basic entities to at least one industry solution entity;
f) selecting a specific subset of basic entities relevant for a specific customer;
g) assigning to at least one basic entity of the specific basic entity subset a basic value representing the relevance of the basic entity for the specific customer;
h) generating a specific subset of industry solution entities as a result of the selected basic entities in their relevant relationship network;
i) selecting relevant industry solution entities from the specific solution entity subset;
j) generating a specific diagram of industry solutions based on the selected relevant industry solution entities of the previous step.

Embodiments may further include calculating a ranking value for prioritizing the industry solutions, wherein the calculation is based on the basic values assigned to the basic entities in their relevant relationship network.

Embodiments may preferably include updating the repository with the gathered information.

In embodiments, the updating may conveniently include applying machine learning techniques.

Furthermore, a computer program element can be provided, comprising computer program code for performing steps according to the above mentioned method when loaded in a digital processor of a computing device.

Additionally, a computer program product stored on a computer usable medium can be provided, comprising computer readable program code for causing a computing device to perform the mentioned method.

Embodiments enable obtaining solution replications leading to reduced costs and shorter times to product delivery for industry suppliers.

Embodiments enable a solution diagram consistent with a specific industry and fitting specific needs of a specific customer.

Embodiments enable a seamless process for customer proposal, from the initial phase of "discovery" to the delivery and deployment phases.

Embodiments provide a solution diagram based on a structured and specific knowledge base.

Embodiments enable prescriptive selling and consulting.

Embodiments enable global information reuse, consistency and curation.

The invention will now be described in preferred but not exclusive embodiments with reference to the accompanying drawings, wherein:

Figure 1 is a drawing schematically illustrating a model of a software tool in accordance with embodiments.

Figure 2 is a screen shot of a TIPS selection page of a GUI of a Value Modeler illustrating TIP maps in a first exemplary embodiment.

Figure 3 is a screen shot of a TIPS selection page of a GUI of a Value Modeler illustrating the relationship network in a highlighted TIPS map in a second exemplary embodiment.

Figure 4 is a diagram schematically representing the relationships among entities of the value modeler and of a centralized repository in accordance with embodiments.

At least some embodiments of the present invention address the above described issue in which a process or system is generating a specific diagram of industry solutions.

A repository is provided. The repository may preferably include industry specific information. The repository comprises a set of basic entities ("TIP") representing a set of industry specific indicators and a set of industry solution entities representing enablers of process improvements. At least two basic entities are linked to generate at least one relationship network. Each relationship network of basic entities is linked to at least one industry solution entity.

A specific subset of basic entities relevant for a specific customer is selected.

At least one basic entity of the specific entity subset is assigned a basic value, called business relevance ("BR"), representing the relevance of the basic entity for a specific customer.

A specific subset of industry solution entities is generated as a result of the selected basic entities in their relevant relationship network.

For the specific customer, the relevant industry solution entities are selected from the specific solution entity subset.

The specific diagram of industry solutions is generated based on the previously selected relevant industry solution entities.

In embodiments, a ranking value for prioritizing the industry solutions may conveniently be calculated based on the basic values assigned to the basic entities in their relevant relationship network.

In embodiments, the repository is updated with the gathered inputted information via a curation process. Preferably, the information updating of the gathered knowledge base includes applying machine learning techniques.

The objects of the first set, i.e. the basic entities, are also called "Trends, Implications and Possibilities" ("TIP") entities.

The objects of the second set, i.e. the industry solution entities, are also called solution templates ("ST").

The TIP basic entities are textual objects with descriptors characterizing an industry, e.g. aerospace, automotive, food & beverage and other industries. Hence, the TIP basic entity may be seen as a textual description, and/or one or more tags to be associated to one or more industry.

The solution template objects are enablers of process improvements linked to product items of the product portfolio of an industry supplier.
The TIP basic entities of the first set are interrelated with each other in a relationship network, also called TIPS map or TIPS path.

The customer requirements may advantageously be guided also by the relationships existing among TIP entities (for example, an Implication object may be correlated to one or more Trends and one or more Possibilities).

The industry solution entities of the second set are the result of the selection performed on the entities of the first set and on their underlying mutual relationships. The underlying mutual relationships are defined by linking at least two TIP basic entities in a relationship network. Each relationship network is linked to at least one industry solution entity.

Thus, advantageously, the industry solutions relevant for a specific customer are automatically generated as a result of the network of relationships of the selected TIP entities.

In embodiments, a ranking value for a Solution Template, representing the business relevance of a solution, may be calculated based on the basic values of the entities in the relevant relationship network. Advantageously, the ranking values of the solution templates enable to rank them according to a prioritization scale so that for example, the user, starting from a set of "suggested" solutions, can select some of them in accordance with their ranking values.

Figure 1 is a drawing schematically illustrating a model of a software tool in accordance with embodiments.

In the exemplary embodiment of Figure 1 it is shown a schematic model 100 of a software tool, herein called Value Modeler, which may preferably be a web based tool. The value modeler 100 connects the sales based TIP framework 101 to the Solutions 102 underpinned by the product portfolio of an industry solution supplier.

In embodiments, the Value Modeler is the enabler for connecting the Value on the "Sales" side to the Best Practices on the "Services" side, allowing to automatically identify the appropriate and consistent set of Solutions 102. On the right side of the Value Modeler Figure 1, the shown steps Delivery, Adoption and Realization 103 represent the steps performed by the solution delivery team of the industry supplier to meet the specific customer requirements.

Figure 2 is a screen shot of a TIPS selection page of a GUI of a Value Modeler illustrating TIP maps in a first exemplary embodiment.

Figure 3 is a screen shot of a TIPS selection page of a GUI of a Value Modeler illustrating the relationship network in a highlighted TIPS map in a second exemplary embodiment.

The TIPS maps shown in Figure 2 and 3 advantageously enable the user to have a guided path in the selection of the relevant TIP entities 201 and relevant industry solutions 202.

As shown in Figure 2, a subset of the TIP basic entities 201 is selected as "relevant", marked with a "V", for the specific customer. Then it is possible to assign a basic value (not shown) to each basic entity of the selected subset.

The set of the solution templates is generated based on the selected TIP entities, and eventually their calculated ranking values based on the assigned basic values, is advantageously provided to the user. The user is conveniently enabled to select the relevant solution templates 202 from this set, also with the help provided by the rankings. In embodiments, the user may select other solutions, even without selecting any TIP, and assign them a ranking value.

In embodiments, pursuit-specific TIPS may also be added from the TIPS selection page of the value modeler.

The customer selection of Solution Template and of Solution Process Improvements, the value side of the solutions, are used as indication of specific customer needs and, in embodiments, may be stored in a repository and become part of the knowledge base for updating the repository.

The network of relationship 301 in the TIPS map is highlighted (dashed line) in the screen shot of Figure 3.

In the GUI it is also present a Prioritization page 203, 303, showing the business relevance rankings of the solution templates ("ST").

To each selected TIP basic entity a BR value is assigned in accordance with the relevance of the basic entity for the given customer.

In embodiment, the basic value BR may for example be a weight value in the range from 1 to 5, where the higher score represents the higher relevance of the entity/item in the customer value network, e.g. the meaning of each ranking is: *1: "Very Low", 2: "Low", 3: "Average", 4: "High", 5: "Very High"*.

In embodiments, the business relevance of each industry solution entity is calculated from the business relevance of its parent objects, i.e. the TIP entities of the corresponding relationship network.

In embodiments, upon need, the user may "override" the calculated business relevance, to compensate for a missing business relevance or to directly set the value. For example, this may happen whenever no parent object is selected as "relevant" (in this case is not possible to apply the calculation model).

In embodiments, the calculation engine may run on the server-side and may be invoked at the assignment of the business

relevance for the basic TIP entities, automatically updating the business relevance of the related solution template.

Given the basic values BR of the basic TIP subset, a ranking value BR of the industry solution entities may be computed, in embodiments, with the formula F1 below, by averaging the BRs of the relevant TIP that belong to the ST network; where $BRT_j$, $BRI_n$, $BRP_k$ are the BR of the TIP entities marked as relevant for the customer pursuit and linked to the $ST_i$.

(F1) $$BRST_i = \frac{\sum_1^j BRT_j + \sum_1^n BRI_n + \sum_1^k BRP_k}{j+n+k}$$

Where in the above formula F1:
$BRT_j$ is the business relevance of the selected *Trend* $T_j$, belonging to the path/relationship network of $ST_i$, where $j$ *is* the number of selected Trends belonging to the path of $ST_i$ (only the ones which have an assigned business relevance).

$BRI_n$ is the business relevance of the selected Implication $I_n$, belonging to the path of $ST_i$, where $n$ is the number of selected Implications belonging to the path of $ST_i$ (only the ones which have an assigned business relevance).

$BRP_k$ is the business relevance of the selected Possibility $P_k$, belonging to the path of $ST_i$, where $k$ is the number of selected Possibilities belonging to the path of $ST_i$ (only the ones which have an assigned business relevance).

The industry solution *ST* in scope for the Pursuit may thus be conveniently ranked in respect of calculated Business Relevance and provided to the customer.

In embodiments, the Business Relevance calculated for the ST may for example be a weight in the range from 1 to 5, where the higher score represents the higher relevance of such processes for the customer. For example, the meaning of each ranking may be *1: "It is a secondary process that is impacting very little the result of the Customer's activities"; 2: "Improvement to this process may bring some limited value in individual Business Domains"; 3: "It is a process that can bring significant benefits in some Customer's activities but not cross domain critical"; "Impacting multiple business areas and key processes where substantial benefits will result 5: "This topic is mission critical with the possibility to massively impact the Company KPIs".*

The generated roadmap is tightly connected to the product portfolio of the solutions of the industry supplier, and can be finally uploaded to the centralized pursuits repository (the same where are created the Industry specific catalogs). The curation phase is an additional optional step to enrich the catalogue.

Figure 4 is a diagram schematically representing the relationships among entities of the value modeler and of a centralized repository in accordance with embodiments.

In the centralized product repository 401 there are object entities of a catalog 403 which are industry specific like TIPS, Collateral, Big Block and Service Content.

In the Value Modeler 402 there are object entities of the customer specific entity model called pursuit 404.

Once a pursuit 404 is created in the Value Modeler, it is possible to create a solution roadmap starting from the data obtained by inheritage 405 from the underlying catalogs of the industry specific to that customer.

The initial set of data can be extended adding customer specific information (TIPS, Process characteristics, KPIs, notes), which are valid only in the pursuit context.

The catalog provides a set of industry specific knowledge base metrics and entities, as for example for industries like pharma or automotive. Each catalog may contain TIPS, Metrics, Solution Process Improvements, Solution Templates objects and others.

Each Solution Template object is associated to one or more industry solutions, software and/or hardware ones.

In embodiments, the catalogs used in the pursuit of the Value Modeler may come from the centralized repository.

The Value Modeler allows managing the pursuits of specific customers, starting from the set of industry specific knowledge based catalogues.

A curation process 406 is in place for updating the catalogs. Advantageously, the gathered information is reused in order to extend and improve the Industry Solutions.

Thus, once the pursuit is uploaded, the curation process 401 may optionally occur to conveniently “promote” customer specific information to a new version of the industry

specific catalogs. Hence, the Value Modeler automatically produces all the customer proposal data, improving efficiency and consistency.
In particular, the Value Modeler automatically generates the "Big Block" object departing from the pursuit that represents the identified solution diagram for the specific customer of the specific selected industry. In such a way, the specific diagram proves consistent with the generic industry specific solution product portfolio of an industry supplier.
In embodiments, such solution diagram may then extended and customized according to further customer requirements.

In embodiments, a specific solution diagram of industrial software modules may be automatically generated by:
- providing a centralized repository with a set of generic catalogue entities comprising generic basic entities and solution template entities linked to the basic entities;
- defining an underlying relationship among solution template entities and a set of software modules;
- upon selection of one catalogue entity, generating an inheriting specific pursuit entity;
- selecting a subset of basic entities from the specific pursuit entity;
- generating a specific solution diagram of software modules corresponding to the software modules having the inherited underlying relationships to the solution template entities linked to the selected basic entities.

**Claims**

1. Method for generating a specific diagram of industry solutions, comprising the following steps:

a) providing a repository (401);

b) providing, within the repository (401), a set of basic entities (201) representing a set of industry specific indicators;

c) providing, within the repository (401), a set of industry solution entities (202) representing enablers of process improvements;

d) linking at least two basic entities (201) to generate at least one relationship network (301);

e) linking the at least one relationship network of basic entities to at least one industry solution entity (202);

f) selecting a specific subset of basic entities (201) relevant for a specific customer;

g) assigning to at least one basic entity (201) of the specific basic entity subset a basic value representing the relevance of the basic entity for the specific customer;

h) generating a specific subset of industry solution entities (202) as a result of the selected basic entities in their relevant relationship network;

i) selecting relevant industry solution entities from the specific solution entity subset;

j) generating a specific diagram of industry solutions based on the selected relevant industry solution entities of the previous step.

2. The method of claim 1, wherein step h) further includes calculating a ranking value for prioritizing the industry solutions, wherein the calculation is based on the basic

values assigned to the basic entities in their relevant relationship network.

3. The method of claim 1 or 2, further including updating the repository with the information gathered in any of the steps d) to j).

4. The method of claim 3, wherein the updating includes applying machine learning techniques.

5. The method according to any of the previous characterized in that it is implemented in software.

6. A system for generating a specific diagram of industry solutions, comprising:

a) a repository;
b) said repository (401) comprising a set of basic entities (201) representing a set of industry specific indicators;
c) said repository (401) comprising a set of industry solution entities (202) representing enablers of process improvements;
d) means for linking at least two basic entities (201) to generate at least one relationship network (301);
e) means for linking the at least one relationship network of basic entities to at least one industry solution entity (202);
f) means for selecting a specific subset of basic entities (TIP) relevant for a specific customer;
g) means for assigning to at least one basic entity (201) of the specific basic entity subset a basic value representing the relevance of the basic entity for the specific customer;

h) means for generating a specific subset of industry solution entities as a result of the selected basic entities in their relevant relationship network;
i) means for selecting relevant industry solution entities from the specific solution entity subset;
j) means for generating a specific diagram of industry solutions based on the selected relevant industry solution entities of the previous step.

7. The system of claim 6, wherein items h) further includes means for calculating a ranking value for prioritizing the industry solutions, wherein the calculation is based on the basic values assigned to the basic entities in their relevant relationship network.

8. The system of claim 6 or 7, further including means for updating the repository with the information gathered in any of the items d) to j).

9. The system of claim 8, further including means for applying machine learning techniques.

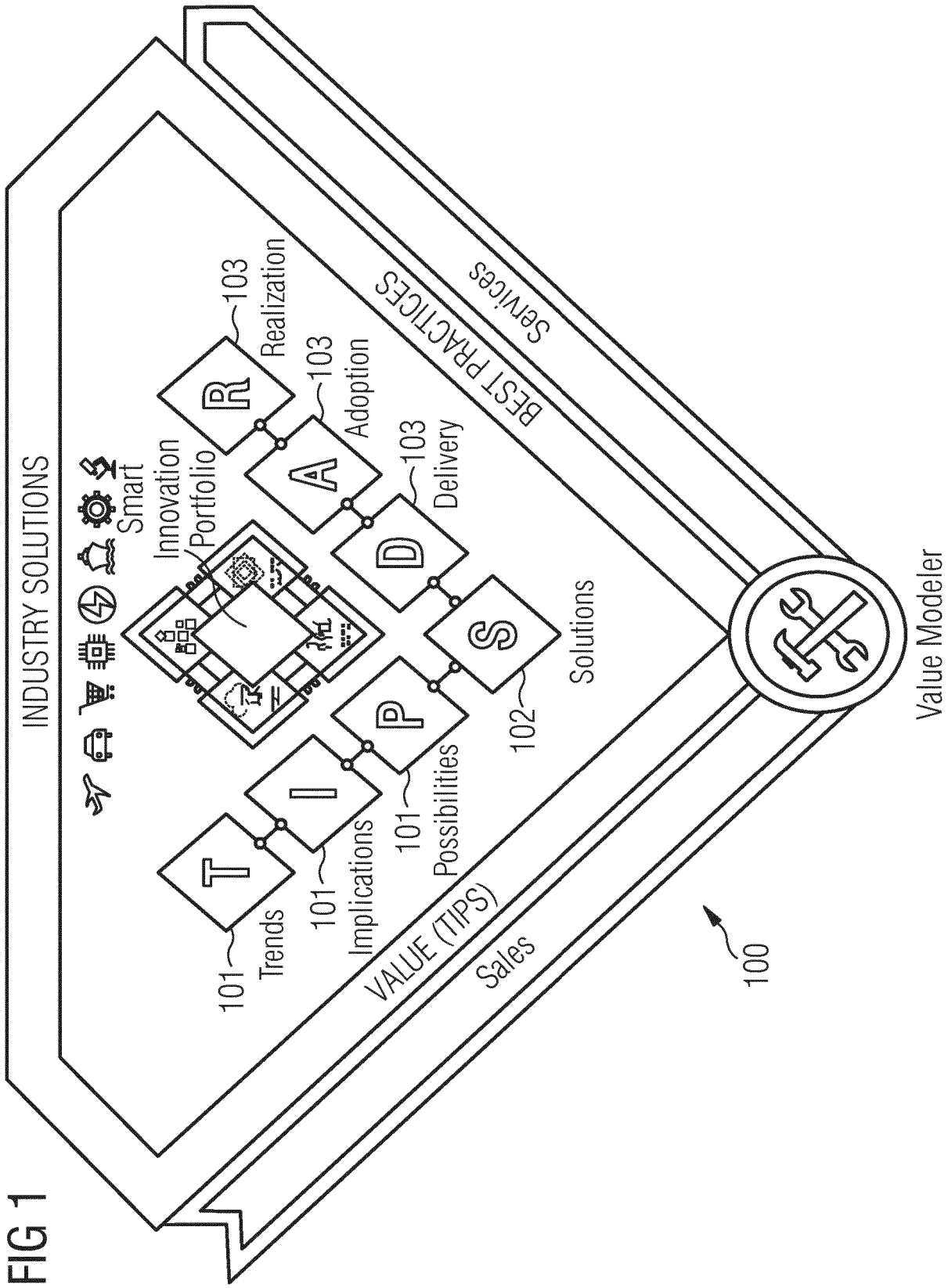

INDUSTRY SOLUTIONS
Smart
Innovation Portfolio
R
103
Realization
A
103
Adoption
D
103
Delivery
S
Solutions
102
P
101
Possibilities
I
101
Implications
T
101
Trends
BEST PRACTICES
Services
VALUE (TIPS)
Sales
Value Modeler
100

FIG 1

FIG 2

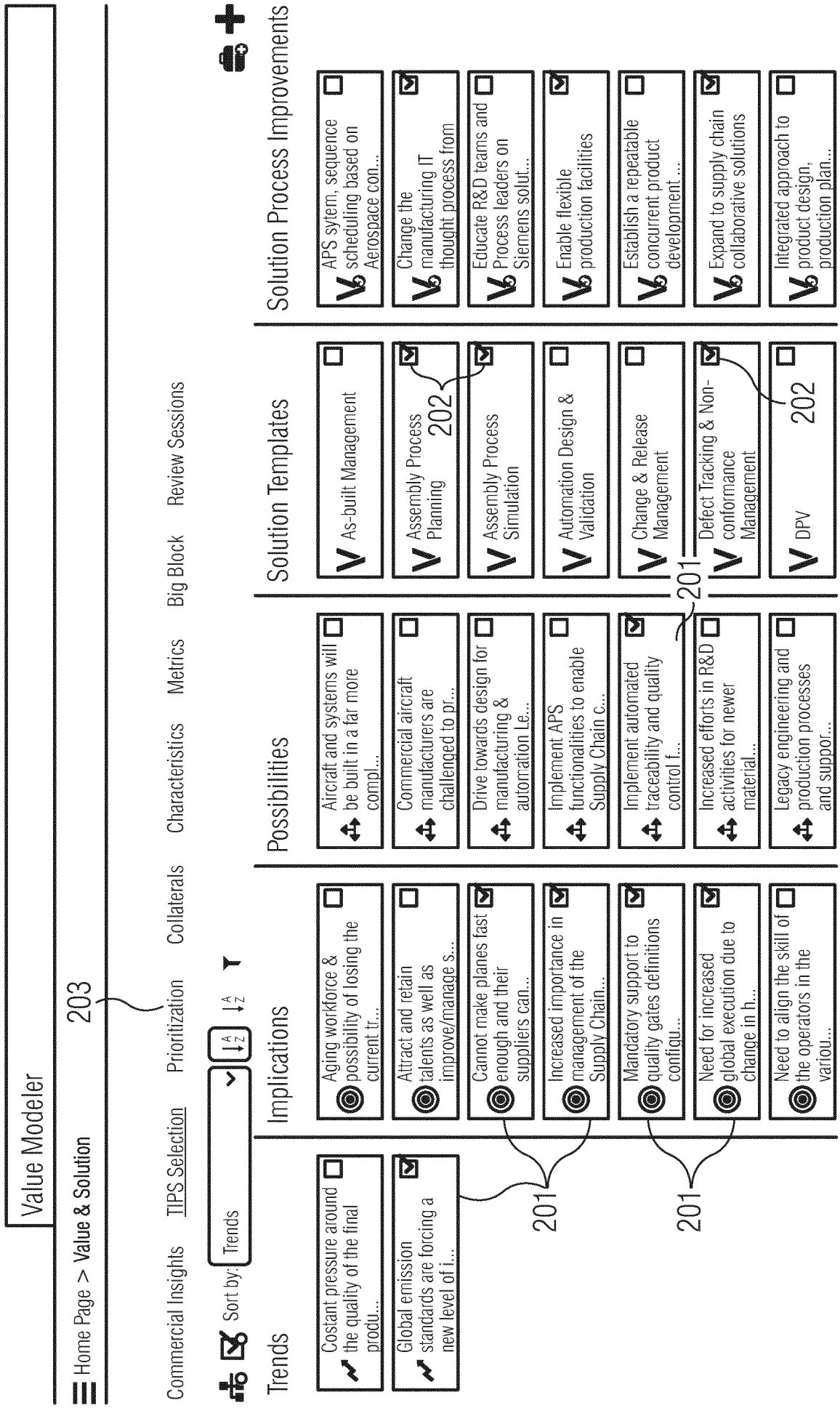
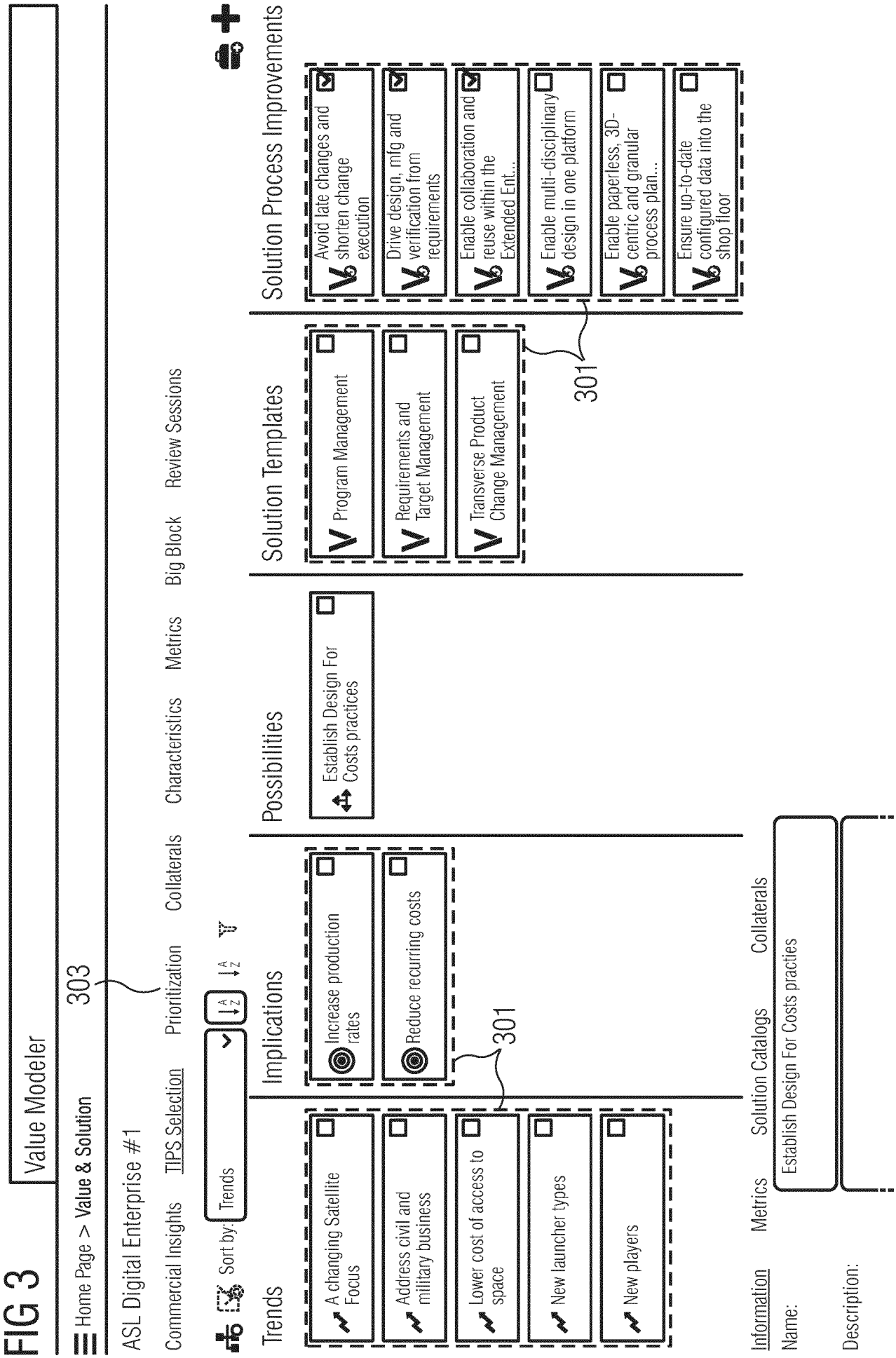

Value Modeler
Home Page > Value & Solution
203
Commercial Insights
TIPS Selection
Prioritization
Collaterals
Characteristics
Metrics
Big Block
Review Sessions
Sort by: Trends
Trends
Implications
Possibilities
Solution Templates
Solution Process Improvements
Costant pressure around the quality of the final produ...
Global emission standards are forcing a new level of i...
Aging workforce & possibility of losing the current tr...
Attract and retain talents as well as improve/manage s...
Cannot make planes fast enough and their suppliers can...
Increased importance in management of the Supply Chain...
Mandatory support to quality gates definitions configu...
Need for increased global execution due to change in h...
Need to align the skill of the operators in the variou...
Aircraft and systems will be built in a far more compl...
Commercial aircraft manufacturers are challenged to pr...
Drive towards design for manufacturing & automation Le...
Implement APS functionalities to enable Supply Chain c...
Implement automated traceability and quality control f...
Increased efforts in R&D activities for newer material...
Legacy engineering and production processes and suppor...
As-built Management
Assembly Process Planning
Assembly Process Simulation
Automation Design & Validation
Change & Release Management
Defect Tracking & Non-conformance Management
DPV
APS sytem, sequence scheduling based on Aerospace con...
Change the manufacturing IT thought process from
Educate R&D teams and Process leaders on Siemens solut...
Enable flexible production facilities
Establish a repeatable concurrent product development ...
Expand to supply chain collaborative solutions
Integrated approach to product design, production plan...
201
202
201
201
202

FIG 3


Value Modeler
Home Page > Value & Solution
303
ASL Digital Enterprise #1
Commercial Insights
TIPS Selection
Prioritization
Collaterals
Characteristics
Metrics
Big Block
Review Sessions
Sort by:
Trends
Trends
A changing Satellite Focus
Address civil and military business
Lower cost of access to space
New launcher types
New players
Implications
Increase production rates
Reduce recurring costs
301
Possibilities
Establish Design For Costs practices
Solution Templates
Program Management
Requirements and Target Management
Transverse Product Change Management
301
Solution Process Improvements
Avoid late changes and shorten change execution
Drive design, mfg and verification from requirements
Enable collaboration and reuse within the Extended Ent...
Enable multi-disciplinary design in one platform
Enable paperless, 3D-centric and granular process plan...
Ensure up-to-date configured data into the shop floor
Information
Metrics
Solution Catalogs
Collaterals
Name:
Establish Design For Costs practies
Description:

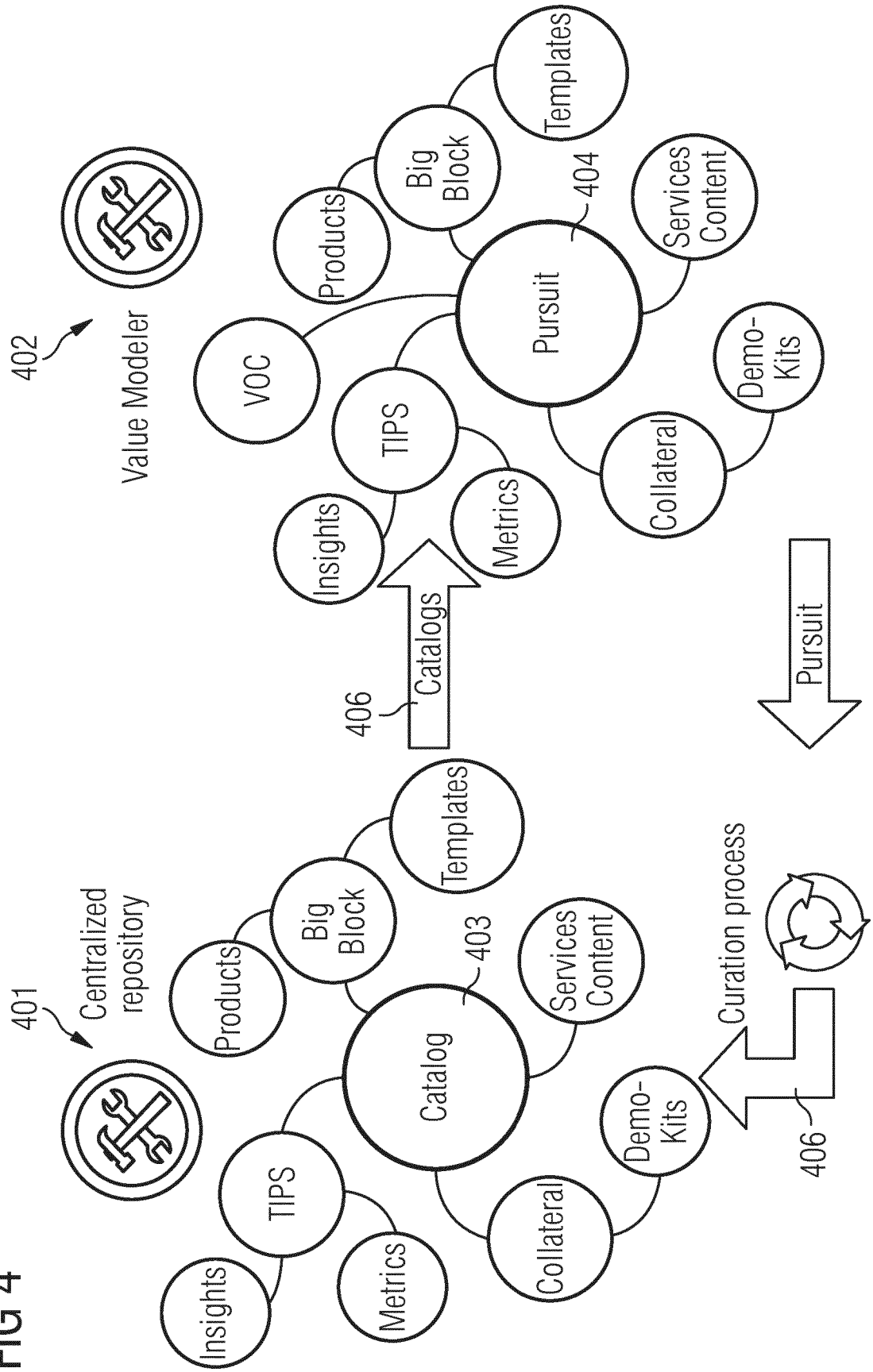

402
Value Modeler
VOC
Products
Big Block
Templates
Pursuit
404
Services Content
TIPS
Insights
Metrics
Collateral
Demo-Kits
406
Catalogs
Pursuit
401
Centralized repository
Insights
TIPS
Metrics
Products
Big Block
Templates
Catalog
403
Services Content
Collateral
Demo-Kits
Curation process
406

FIG 4

# INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2017/071961

**A. CLASSIFICATION OF SUBJECT MATTER**
INV. G06Q50/04
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

**B. FIELDS SEARCHED**

Minimum documentation searched (classification system followed by classification symbols)
G06Q

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

**C. DOCUMENTS CONSIDERED TO BE RELEVANT**

| Category* | Citation of document, with indication, where appropriate, of the relevant passages | Relevant to claim No. |
|---|---|---|
| X | US 2008/162266 A1 (GRIESSMANN JENS [DE] ET AL) 3 July 2008 (2008-07-03)<br>abstract<br>figures 1-4<br>paragraph [0001] - paragraph [0005]<br>paragraph [0011] - paragraph [0069]<br>claims 1-34<br>----- | 1-9 |
| X | US 2007/179793 A1 (BAGCHI SUGATO [US] ET AL) 2 August 2007 (2007-08-02)<br>abstract<br>figures 4-10<br>paragraph [0002] - paragraph [0010]<br>paragraph [0022] - paragraph [0040]<br>claims 1-9<br>----- | 1-9 |
| | -/-- | |

[X] Further documents are listed in the continuation of Box C.

[X] See patent family annex.

* Special categories of cited documents :

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"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

| Date of the actual completion of the international search | Date of mailing of the international search report |
|---|---|
| 21 September 2017 | 09/10/2017 |

| Name and mailing address of the ISA/ | Authorized officer |
|---|---|
| European Patent Office, P.B. 5818 Patentlaan 2<br>NL - 2280 HV Rijswijk<br>Tel. (+31-70) 340-2040,<br>Fax: (+31-70) 340-3016 | Bassanini, Anna |

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# INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2017/071961

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

| Category* | Citation of document, with indication, where appropriate, of the relevant passages | Relevant to claim No. |
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# INTERNATIONAL SEARCH REPORT

**Information on patent family members**

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